



MAT110

Differential Calculus and Coordinate Geometry

Assignment 01

Spring 24

Submission Date: 14th February, 2024 (Wednesday)

Instructions:

Assignment should be handwritten. Please write your Name and ID on the cover page of your assignment answer script. Make PDF of your assignment and submit it to the provided google form.

Solve all problems. Answer the questions by yourself. Plagiarism will lead to an F grade in the course. Each question carries 10 marks. Total marks is 60 and it will be converted to 15.

1. Find the slope and tangent line equations for the given function at the given value of x :

$$f(x) = x + \frac{9}{x}$$

at $x = -3$

2. Find the differential coefficients of

$$(\sin x)^{\cos x} + (\cos x)^{\sin x}$$

3. Prove that,

$$\lim_{x \rightarrow 0} (x + e^x)^{\frac{1}{x}} = e^2$$

4. Find the differential coefficient of $\tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$ with respect to $\sec^{-1} \left(\frac{1}{2x^2-1} \right)$.

5. Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin xy$.

6. Let,

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ \sqrt{x}, & x > 1 \end{cases}$$

Is $f(x)$ continuous at $x = 1$? Determine whether $f(x)$ is differentiable at $x = 1$?